The Zero-Variance Fraction

Diagnosing and Budgeting Signal Starvation in Group-Relative RL Post-Training

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The problem in one picture

GRPO trains on **groups** of G completions per prompt; advantages are centred within the group.

If every completion in a group gets the same reward, the group contributes zero gradient.

Two ways it happens:

- all wrong — task too hard
- all right — task *mastered*

The blind spot
The second case is invisible in the reward curve: reward reads *success* exactly when learning has *stopped*.

The instrument

Definition

$$\mathrm{ZVF}_t = \frac{1}{|\mathcal{P}_t|} \sum_p \mathbf{1}\left[\mathrm{Var}\, r = 0 \text{ within the group}\right] \qquad (\text{gradient utilisation} = 1 - \mathrm{ZVF}_t)$$

- Costs nothing: computed from rewards already in hand
- Population form under binary rewards: $h_G(p) = p^G + (1-p)^G$ — a **U-shape** in difficulty
- Must be read **with reward**: ZVF alone aliases mastery with incapacity

Claim contract (what I defend — and what I don't)

Claim 1. ZVF+reward is a cheap, calibrated online diagnostic of zero-advantage regimes.

Claim 2. At matched rollout budget, group size selects *which end* of training starves ($G{=}2$: all-correct wall; $G{=}16$: none).

Explicit non-claims

no predictive/causal power for ZVF · no data-dependent optimal G · no controller-efficacy claim · no generalisation beyond Qwen3-8B / GSM8K-family / one managed LoRA stack, 1–3 seeds

Scope and infrastructure

- One 8B model (Qwen3-8B, LoRA r4) on a managed training API; vLLM eval on 3 GPU backends; 7-library observational corpus (0.6B–671B logged runs)
- Base capability, GSM8K test (n=32/problem, clustered 95% CIs): **pass@1 30.4% — pass@32 91.0%** $\Rightarrow$ little capability headroom; claims scoped accordingly
- Every result: seeded, manifest-ed, W&B-mirrored, resumable checkpoints; partial runs stamped, never promoted

Claim 1 — the all-correct wall (the key exhibit)

Matched budget: **2,560 rollouts per arm**, two seeds each

G=2 × 160 steps vs. G=16 × 20 steps

	G=2	G=16
final train reward	≈ 1.0	comparable
late-run ZVF	0.75–1.0	0.00–0.25
gradient signal at end	none	sustained

Reward says "succeeding"; ZVF says "training is over." **Only the pair tells the truth.**

Theory 1/3 — calibration (T1)

- ZVF_t is an unbiased binomial-proportion estimator; we ship it with a **Wilson interval — empirical coverage 0.95–0.98 in every tested setting**
- Curriculum-ordered batches collapse *global* coverage (0.08–0.40) but the interval stays calibrated for the *local* estimand (0.944) — a labelling requirement, not an invalidation

Theory 2/3 — the reliability budget (T2)

$$N(\mathrm{ZVF}) = G\left\lceil \frac{\ln\delta}{\ln \mathrm{ZVF}} \right\rceil \quad = \quad \text{the } (1-\delta)\text{-quantile of rollouts to the next informative group}$$

- A **budget**, not a minimum — signal can arrive in the first G rollouts w.p. $1-\mathrm{ZVF}$
- Empirically **exact**: model quantile = observed quantile, ratio 1.00, in all six difficulty strata
- Hardest stratum ($\hat{p}\approx 0.01$): 160 rollouts to *guarantee* one usable gradient at $\delta=0.1$

Theory 3/3 — an honest negative result (T3)

Signal-per-rollout objective: $J(G) = \frac{1}{G}\,\mathbb{E}_{\phi}[(1 - h_G(p))\,p(1-p)]$

- Algebraic identity: $1 - p^G - (1-p)^G = G\,p(1-p)$ at $G \in \{2,3\}$
- $\Rightarrow\ J(2) = J(3) = \mathbb{E}[(p(1-p))^2]$ **for every prior** — argmax always $\{2,3\}$
- Our earlier "theory–experiment agreement" was therefore guaranteed, not a test — found in external review, corrected openly
- Consequence: per-rollout accounting cannot justify adaptive G; richer objectives required

Claim 2 — group size is a schedule

- Static sweeps are inconclusive (non-monotone at fixed steps; optimum drifts to $G{\approx}32$ at fixed token budget)
- The matched-**rollout**-budget view resolves it: small G = more optimiser steps early, starves late; large G = fewer steps, signal survives the endgame
- Contrastive ($G{=}2 \sim G{=}16$) equivalence claims hold on final reward, **not on signal availability** — and decay from 97.6% to 72.7% retention as training scale grows
- Controller: designed, evidence-gapped, **explicitly not claimed** pending compute-matched trials vs static $G{=}16$

Loss form — GRPO vs Dr.GRPO (six uncapped arms)

Qwen3-8B, 1,024-token cap, 30 steps, G=8, **3 seeds per loss**

	length first-5→last-10	late ZVF
GRPO	1004→905, 981→944, 996→900	0.47/0.47/0.55
Dr.GRPO	999→931, 972→902, 1000→878	0.45/0.72/0.70

No length inflation in either loss; no ZVF separation. At this scale the loss-form choice has no observable footprint.

The incident that became a chapter

The unwired flag (2026-07-11)
A six-arm GRPO-vs-Dr.GRPO panel silently trained **one** objective: the `--loss` flag was documented but never wired. No reward, length, or ZVF trace revealed it — only reading the runner did.

- Response: invalidate loudly, preserve artifacts, fix, rerun same day
- The corrected panel reversed one conclusion of the invalid one
- Lesson: **stack identity cannot be certified from plausible-looking outputs**

From incident to standard

Eight-item minimum reportable stack — every item once flipped a result in our own corpus:

1. loss form 2. reference policy/KL 3. sampler/backend/precision + **base-checkpoint identity** 4. per-step ZVF trajectory
5. group-size schedule 6. held-out split 7. decontamination + parser probe 8. **pass@k curves**, not pass@1 alone

Documented levers: backend swap $\Rightarrow$ **17**$\times$ reward span; same "DAPO" label $\Rightarrow$ ZVF 0.00 vs 0.58; parser micro-jitter $\Rightarrow$ ZVF 0.158 $\rightarrow$ 0.000. Machine-readable registry + flip-risk `stackdiff` ship with the bench.

Limitations and roadmap

Limitations — one model, one task family, one closed stack, 1–3 seeds; theory has declared proof gaps; ZVF is binary-reward-specific.

Post-degree publication plan (externally reviewed, gated):

- Diagnostic paper (Claims 1–2) — ready without controller claims
- Controller paper — gated on compute-matched win vs static $G{=}16$ *and* reward-only schedules, ≥ 3 seeds
- Survival audit — gated on an open stack + differential-objective test tooling

Conclusion

Under verifiable rewards, group-relative training starves at **both ends** of difficulty.

$\mathrm{ZVF} +$ reward sees both walls — including the one the reward curve cannot show.

Group size decides **which wall** your budget hits.

Thesis, 17 consolidated working papers, and every artifact: `github.com/arvindcr4/tinker-rl-lab`

Backup: provenance of the 17 papers

Benchmark family (4) → Ch.3 · P2+ZVF audit → Ch.4 · theory → Ch.5 · P3+P7+E-R2b → Ch.6 · P4+E-P4 → Ch.7 · P1 → Ch.8 · P5/P6/position/registry/audit → Ch.9 · P8 out-of-programme. Canonicalisation per `PAPERS_README.md`; every claim → artifact in thesis Appendix B.

Backup: key numbers

GSM8K base pass@1/8/32: 30.4/79.7/91.0% (clustered CIs) · train-pool ZVF 10.86%, all all-incorrect · T1 Wilson coverage 0.95–0.98 · T2 fit ratio 1.00 (6 strata) · E-R2b: ZVF 0.75–1.0 vs ≤ 0.25 at 2,560 rollouts · E-P4: lengths shrink 6–12% both losses · MBPP transfer: zero forgetting; pass@32 frontier $+1.5$–$+2.5$ pts (1 seed) · MATH-500: GSM8K gains do not replicate (partial) · held-out GSM8K control: 82.0 $\rightarrow$ 83.3%, p=0.26.

Backup: corrections adopted from review

T2 quantifier error (quantile $\neq$ minimum) · T2 validation direction · T3 universality (J(2)=J(3)) · GU definition · off-by-one final target in 7 trainers · generation seeding · parser v2 (last-boxed) · exact hypergeometric T1 truth. All fixed 2026-07-11 (commits `5486ae2`, `acc08eb`); a further audit pass produced only fabricated findings — all refuted against source, reinforcing the verify-against-the-stack thesis.